

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	Jeyakannan L	Reg. No.:	192525376
Date:	03-8-2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

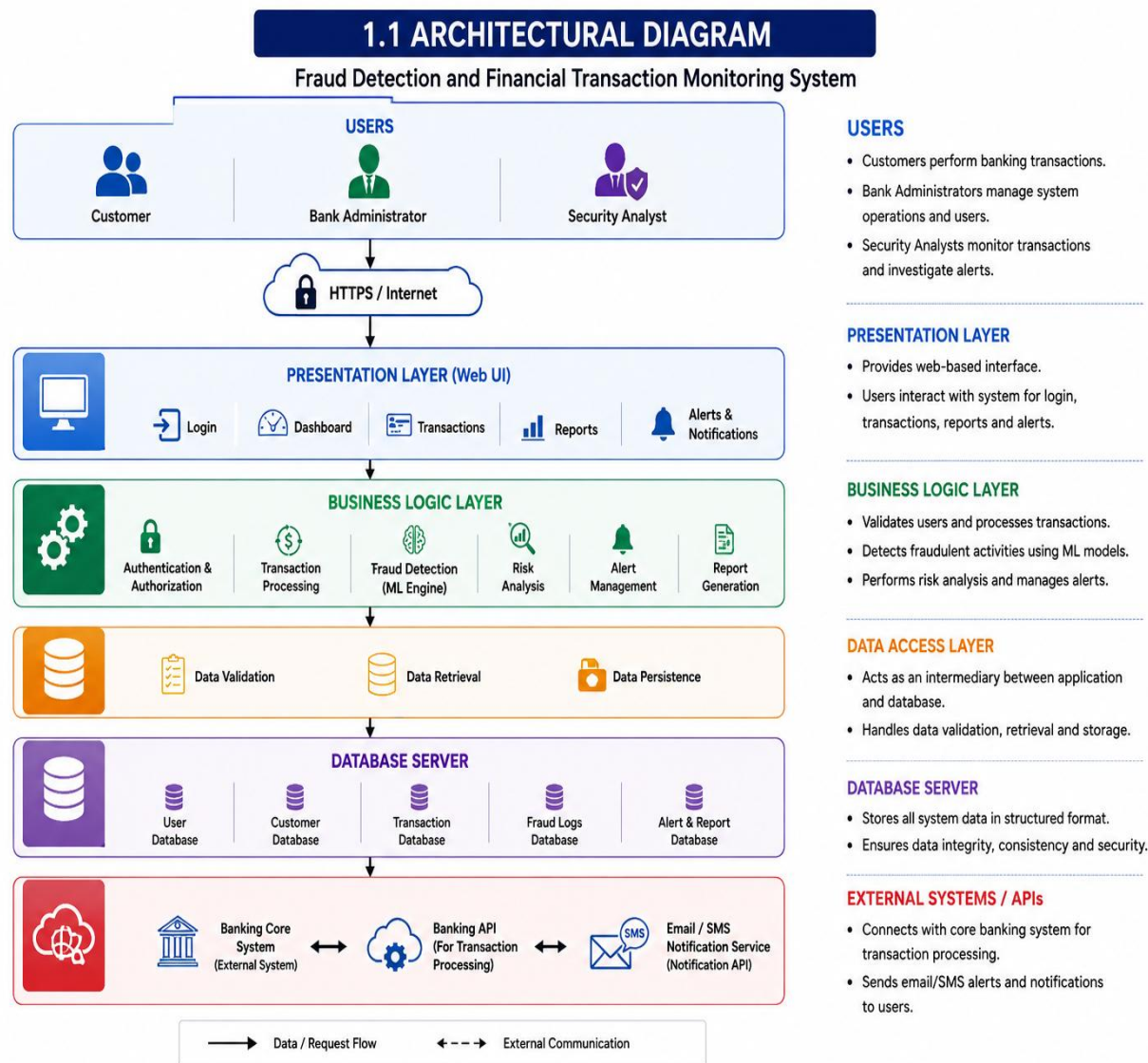
Annexure A – Architecture Design Assignment

1. Analyse System Requirements

1.1 Study the given problem statement and identify the system objectives.

System Objectives

The **Fraud Detection and Financial Transaction Monitoring System** is designed to monitor banking transactions in real time, detect fraudulent activities, reduce financial losses, and improve the security of digital banking services.



1.2 Analyse the functional and non-functional requirements that influence the software architecture.

Functional Requirements

- User authentication and login.
- Process financial transactions.
- Monitor transactions in real time.
- Detect suspicious transactions.
- Generate fraud alerts and reports.

Non-Functional Requirements

- Fast transaction processing.
- Secure authentication and data encryption.
- Reliable fraud detection.
- User-friendly interface.
- High system availability (24×7).

1.3 Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

Quality Attribute	Description
Scalability	Supports increasing users and transaction volume.
Maintainability	Easy to update and maintain the system.
Reliability	Provides accurate and consistent fraud detection.
Availability	Operates continuously with minimal downtime.
Performance	Processes transactions quickly in real time.

2. Select a Suitable Software Architecture

2.1 Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).

Hybrid Architecture (Layered + Client-Server)

The **Fraud Detection and Financial Transaction Monitoring System** uses a **Hybrid Architecture**, combining **Layered Architecture** and **Client-Server Architecture**.

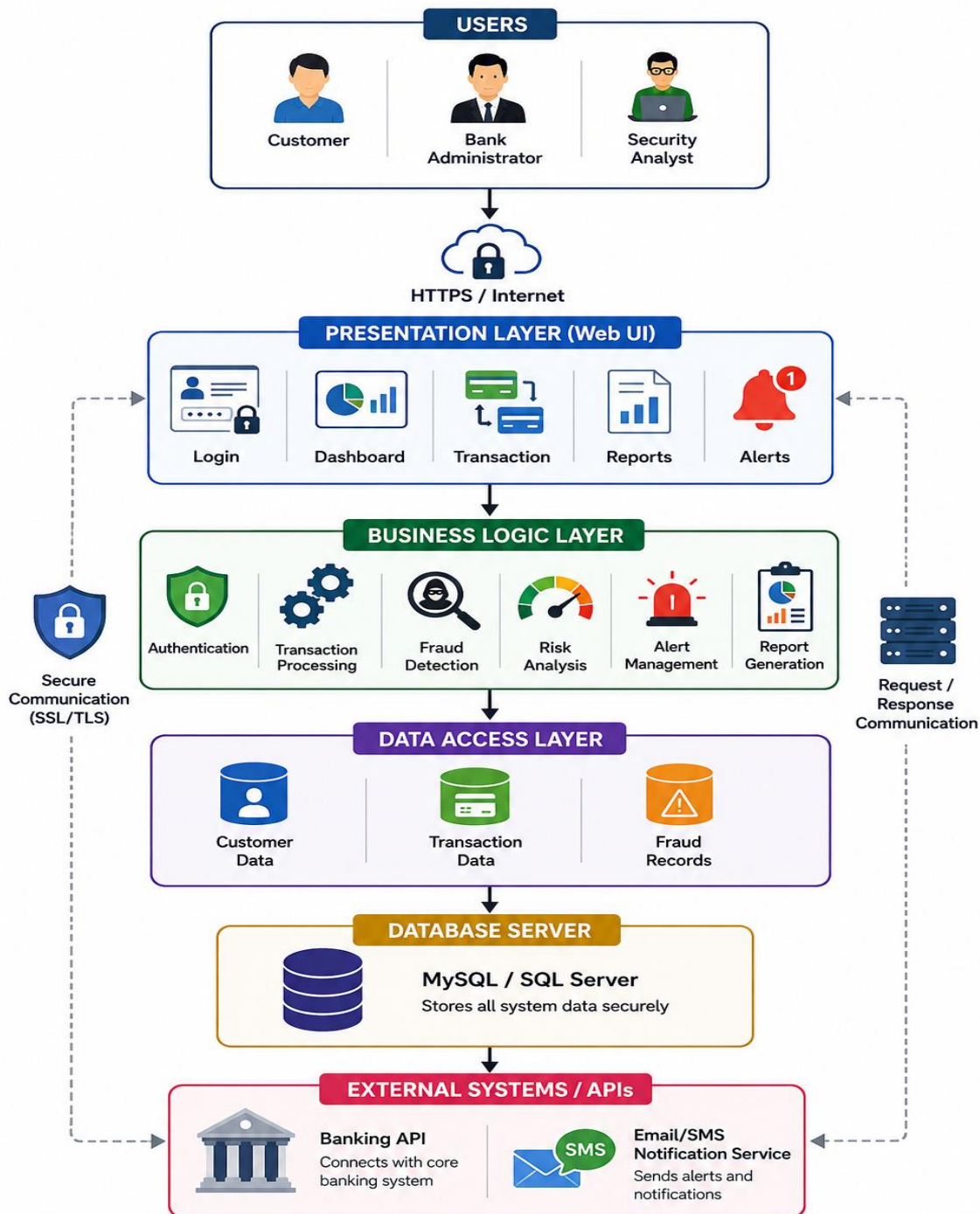
- **Presentation Layer** – User interface for customers and administrators.
- **Business Layer** – Processes transactions and detects fraud.
- **Data Layer** – Stores customer, transaction, and fraud records.
- **Client-Server** – Clients send requests to the central server, which processes and stores data securely.

2.2 Justify why the selected architecture is the most suitable for the given problem.

Reason	Justification
Security	Centralized server protects sensitive banking data.
Scalability	Easily supports more users and transactions.
Maintainability	Layered design makes updates and debugging easier.
Performance	Enables efficient processing of real-time transactions.
Reliability	Ensures stable operation and accurate fraud detection.

3 Draw the Software Architecture Diagram

3.1 Design a clear architecture diagram illustrating the overall structure of the proposed system.



3.2 Include all major layers/components, databases, external systems, APIs, and communication mechanisms.

Component	Description
Presentation Layer	Provides the user interface for Customers, Bank Administrators, and Security Analysts.
Business Logic Layer	Handles authentication, transaction processing, fraud detection, risk analysis, alerts, and report generation.
Data Access Layer	Connects the application with the database and manages data retrieval and storage.
Database Server	Stores customer details, transaction records, fraud logs, and user information (MySQL/SQL Server).
External Systems	Banking Core System, Email/SMS Notification Service.
APIs	Banking API for transaction processing and Notification API for sending alerts.
Communication Mechanism	Secure HTTPS/SSL/TLS protocol for communication between clients, server, database, and external services.
Component	Description

3.3 Use appropriate software architecture notations and labels.



4. Explain Components and Their Interactions

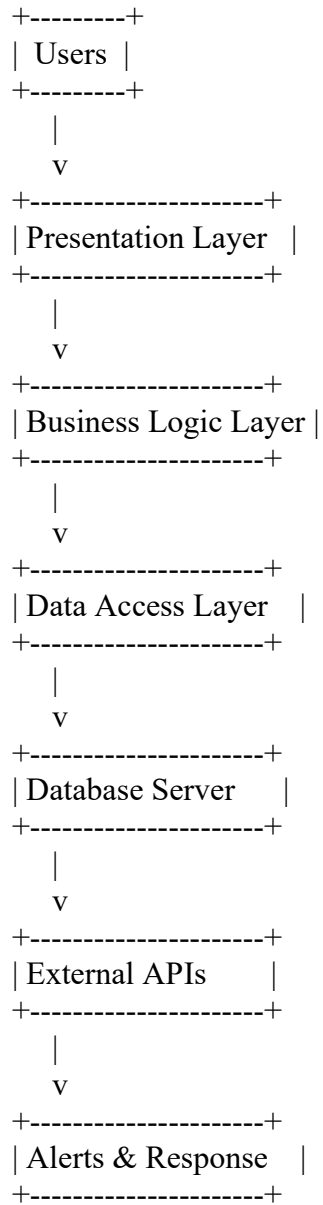
4.1 Describe the purpose and responsibilities of each architectural component.

Component	Purpose / Responsibility
Presentation Layer	Provides the user interface for customers, administrators, and security analysts.
Business Logic Layer	Processes transactions, detects fraud, manages alerts, and generates reports.
Data Access Layer	Retrieves and stores data between the application and database.
Database Server	Stores customer details, transaction records, and fraud logs securely.
External Systems / APIs	Connects with Banking API and Email/SMS services for transactions and notifications.

4.2 Explain how the components communicate and exchange data.

- Users log in through the **Presentation Layer**.
- Requests are sent to the **Business Logic Layer**.
- The **Business Logic Layer** validates transactions and performs fraud detection.
- The **Data Access Layer** retrieves or updates data in the **Database Server**.
- If fraud is detected, the system uses **External APIs** to send Email/SMS alerts.
- The processed result is returned to the user through the **Presentation Layer**.

4.3 Illustrate the overall workflow of the system.



5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

Aspect	Evaluation	Aspect
Scalability	Supports increasing users and transaction volume without affecting performance.	Scalability
Security	Uses secure login, data encryption, HTTPS, and role-based access control to protect sensitive information.	Security
Performance	Processes transactions quickly and detects fraud in real time.	Performance
Reliability	Ensures accurate fraud detection and stable system operation with secure data storage.	Reliability
Maintainability	Layered architecture allows easy updates, testing, and maintenance of individual components.	Maintainability
Availability	Provides 24×7 system access with backup and recovery support to minimize downtime.	Availability

6. Justify Architectural Decisions

6.1 Explain the rationale behind your architectural choices.

The **Hybrid Architecture (Layered + Client-Server)** was selected because it provides a secure, scalable, and well-structured solution for the **Fraud Detection and Financial Transaction Monitoring System**. It separates the application into different layers, making development, testing, and maintenance easier while ensuring efficient transaction processing.

6.2 Discuss the trade-offs considered while selecting the architecture

- High security through centralized data management.
- Easy to maintain and update individual layers.
- Supports increasing users and transaction volumes.
- Improves code organization and system reliability.

6.3 Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

- Easily integrates **AI/ML-based fraud detection** in the future.
- Supports integration with **Banking APIs** and third-party services.
- New modules and features can be added without affecting existing components.
- Modular design simplifies maintenance, testing, and future upgrades.
- Suitable for long-term system growth and expansion.